



UNIVERSITÀ DEGLI STUDI DI GENOVA

DIBRIS

DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY,
BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

COMPUTER VISION

Lab 6
Fundamental matrix estimation

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1 Part I

1.1 8-point algorithm function (version 1)

In order to implement this algorithm, it is essential to initially construct the "A" matrix for the linear homogeneous system. To accomplish this, the `EightPointsAlgorithm(P1, P2)` function is invoked with the corresponding points $P1$ and $P2$. Subsequently, the Singular Value Decomposition ($SVD(A) = [U \ D \ V]$) is employed to produce the unit matrix V and to identify the final column of V as the solution, which is a 9×1 vector. For the F matrix to be constructed, the subsequent step is to restructure the 9×1 matrix into a 3×3 matrix using `reshape( , [3 3])`. The matrix F must have rank 2, and we manually achieve this. Using the $SVD(F) = [U \ D \ V]$, we then set $D[3,3]$ of the matrix D to 0. The F matrix is subsequently recomputed using UDV^T , in which $D[3,3]$ is equal to zero.

1.2 8-point algorithm function (version2)

The normalized eight-point algorithm is employed to improve the accuracy of the fundamental matrix estimation. Initially, the input points are normalized, and the eight-point algorithm from the final section is implemented. The previous algorithm's stages will be executed once more on the normalized corresponding points in the `EightPointsAlgorithmN(P1, P2)`. In order to emphasize the point, $P1$ and $P2$ must be normalized using the `normalise2dpts(P1)` function. Subsequently, the new F is generated by denormalizing the functionality matrix F using the `EightPointsAlgorithm(P1, P2)`.

It is imperative to emphasize that all of the points are presented in a uniform manner. The numerical stability of the solution can be enhanced by normalizing the input points, which is involved in the solution of a linear system of equations in the 8-point algorithm. It aids in the prevention of problems associated with ill-conditioned matrices that may occur when dealing with large or small numbers in the input data. The algorithm becomes less susceptible to anomalies and noise in the data by establishing a common scale for the point coordinates. The fundamental matrix is defined to scale. The algorithm's scale-invariance is guaranteed by normalizing the coordinates. Normalization facilitates the optimization process by facilitating the attainment of more dependable and expedited convergence. It can enhance the solution's robustness, particularly in situations where the point correspondences exhibit a broad spectrum of scales.

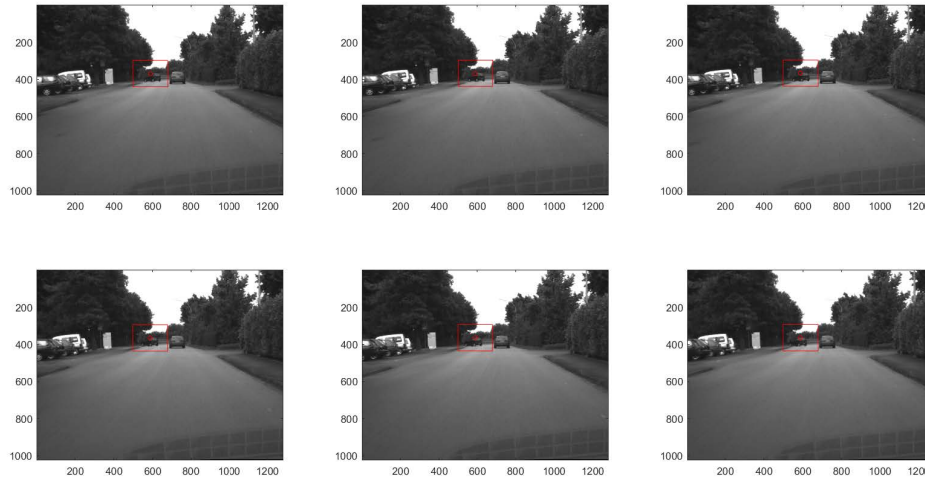


Figure 1: The Second results of detecting objects with larger frames

2 Harris corner detection

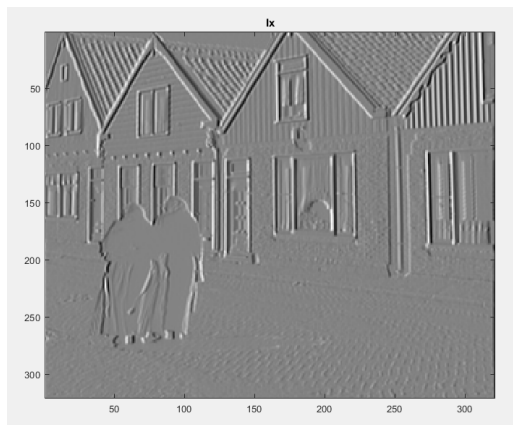
The Harris Corner Detector is a well-known algorithm that is utilized for the purpose of locating corners or spots of interest within an image. The primary objective of the Harris Corner Detector is to identify substantial variations in intensity that occur in a variety of directions within a local neighborhood of a picture. There are points in the image when the intensity of the image suddenly changes in many directions, and these are known as corners. A structure tensor, which is a matrix that summarizes the local picture structure based on the image derivatives, might be constructed for each pixel in the image. This would allow us to construct a structure tensor. This is the formula for the structural tensor:

$$M = \sum_{x,y} W(x,y) \begin{bmatrix} I_x I_x & I_x I_y \\ I_x I_y & I_y I_y \end{bmatrix}$$

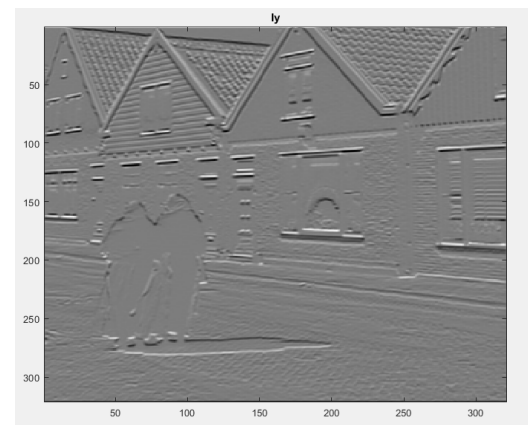
Utilizing the determinant and trace of the structure tensor, the following steps can be taken to compute a corner response function:

$$R = \det(M) - \alpha \cdot \text{trace}^2(M)$$

Additionally, the parameter α is a constant that is used to alter the sensitivity of the detector, which typically falls between the range of 0.04 to 0.06. When it comes to this particular exercise, the value that is deemed to be between these two numbers is 0.05. When analyzing the image, the corners are determined by the scores that are more than 0.3 of the maximum value of the R matrix. By adjusting this value, we are able to determine the total number of points that can be classified as the corner. The partial derivatives of the image are depicted in Figure 9, and the Gaussian filter is depicted in Figure 10.



(a)



(b)

Figure 2: The partial derivatives of the image

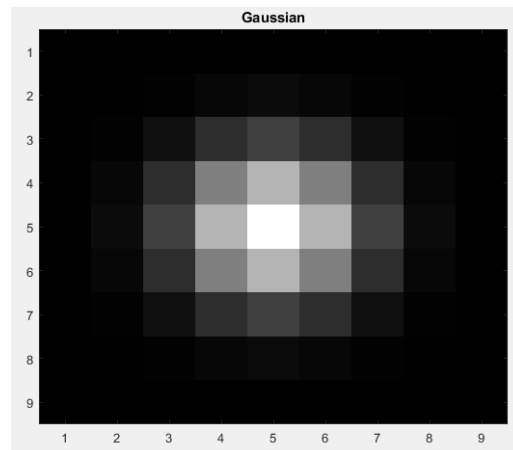


Figure 3: The Gaussian filter



Figure 4: The detected corners

There are three functions inside of the files that have been provided for the assignment. "Template Matching" is the name of the function that is responsible for fitting templates together. The function known as "Color-Based Segmentation" also performs a simulation of the color-based segmentation approach on the displayed images. The corner detection functionality is applied to the image by utilizing the Harris Corner() function. Every one of these routines is written in a separate file that is referred to as "main." You have the option of commenting out each function in the event that you do not require the results pertaining to that particular aspect of the problem.